

Investor Composition, Bond Liquidity, and Labor Market

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Preliminary

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Motivation

Two important aspects of the corporate bond market:

- Bond market liquidity has real effects on firms, including employment.
- Bond investor base has shifted dramatically: mutual funds' share has roughly tripled over three decades and unlike buy-and-hold insurers, they offer daily redemption and sell into stress.

⇒ **Who holds a firm's bonds** may determine how hard liquidity shocks hit its workers.

This paper asks:

- ❶ Does **investor composition** shape how liquidity risk premium affects the labor market?
- ❷ Can a heterogeneous-firm, heterogeneous-investor model rationalize and quantify it?

This Paper

Empirics: panel of U.S. firms matched to their bondholder base (eMAXX), 2002–2023

- Firms' bond liquidity risk negatively predicts employment growth, investment, debt issuance.
- The effect is **amplified by mutual fund ownership**: moving from 25th to 75th percentile of MF ownership adds **~50%** of the baseline employment effect.
- **Causal**: mandate-induced maturity-cutoff IV + shift-share design.

Placebo: no amplification for insurer ownership — channel is specific to **run-prone** investors.

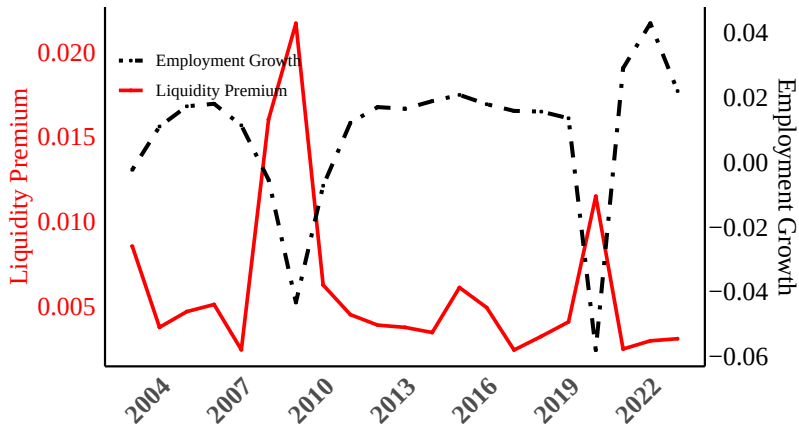
Quantitative Model: heterogeneous firms issue **long-term defaultable bonds** to heterogeneous investors who trade in an **OTC market** with search frictions (micro-found liquidity premia)

- Liquidity reaches the firm through the **primary-market issue price**: issuance proceeds move with bond-market liquidity $\Rightarrow$ hiring, investment, default.
- Investor heterogeneity = **risk capacity** of the marginal buyer: mutual-fund-like (shallow) vs. insurer-like (deep) demand.
- Deep learning based global solution to the model.

Roadmap

- ➊ Data and Empirical Results
- ➋ Model: Firms and an OTC Bond Market
- ➌ Conclusion

Liquidity Risk and Employment Growth Move Together



Correlation = -0.78 : liquidity dries up exactly when employment collapses — 2008–09, 2020.

vs. Unemployment Rate

Data

Sample: U.S. nonfinancial firms with traded bonds and CDS, 2002–2023, annual

- **Key variable 1 — liquidity premium:** bond yield spread (TRACE) – CDS spread (Markit), à la Longstaff–Mithal–Neis (2005)
 - strips out default compensation $\Rightarrow$ isolates the liquidity component
- **Key variable 2 — bond ownership:** mutual fund and insurer holdings of each firm's bonds, from eMAXX
 - security-level holdings of buy-side institutions, aggregated to the firm
- **Firm outcomes:** employment, investment, debt issuance (Compustat)
- **Other:** macro and labor aggregates (FRED, BLS); fund flow shocks (Dou–Kogan–Wu 2025, WRDS)

Firm Level Panel Regression

Empirical design: does the liquidity–employment link run through *who holds the bonds*?

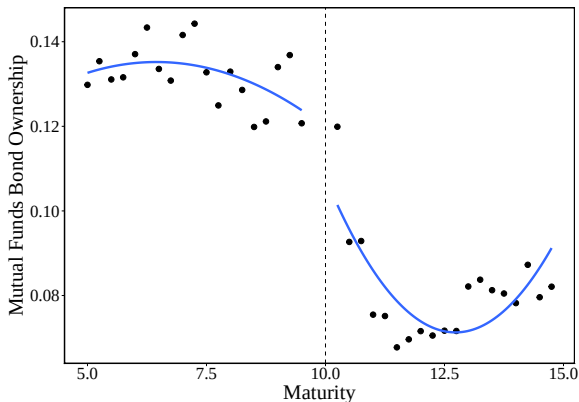
$$\Delta \text{Emp}_{i,t} = \beta_1 \underbrace{\text{Liq. Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1}}_{\text{transmission channel}} + \beta_2 \underbrace{\text{Liq. Premium}_{i,t-1}}_{\text{direct exposure}} + \beta_3 \text{Bond Ownership}_{i,t-1} + X'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

- $\beta_2 < 0$: firm-level analogue of the aggregate fact
- $\beta_1 < 0$: **mutual fund ownership amplifies** transmission — our object of interest
- Identification challenge: ownership is endogenous $\Rightarrow$ mandate-cutoff IV

× Fin. Constraints, Recession

IV Construction: The 10-Year Maturity Cutoff

- Literature documents that a large fraction of corporate bond funds are intermediate-term, and only invest in bonds whose maturity is **less than** 10 years due to investment mandates.
- So bonds with time-to-maturity less than 10 years face **larger demand from mutual funds**.



IV Construction: The Mandate IV (MIV)

- Exploiting this investment-mandate-induced 10-year maturity cutoff, we follow Bretscher, Schmid, Ye (2025) to construct the instrumental variable MIV.

$$MIV_{jt} = \log \left(\sum_{h=1}^2 \frac{A_{ht}}{N_{ht}} \cdot \mathbb{1}_{jth} \right), \quad (1)$$

- A_{ht} : aggregate assets of **mutual funds** whose mandate is maturity category h in month t ,
- N_{ht} : number of bonds outstanding in maturity category h at t
 $\Rightarrow A_{ht}/N_{ht}$ = mutual-fund assets per bond in the bucket.

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 $\Rightarrow A_{ht}/N_{ht}$ = mutual-fund assets per bond in the bucket.
- Aggregate to the firm with **par weights frozen at $t - k$** : firm i 's bonds outstanding at $t - k$, $\mathcal{B}_{i,t-k}$, weighted by their amounts then:

$$\widetilde{MIV}_{it}^{firm} = \frac{\sum_{j \in \mathcal{B}_{i,t-k}} AmtOut_{j,t-k} \cdot MIV_{jt}}{\sum_{j \in \mathcal{B}_{i,t-k}} AmtOut_{j,t-k}}, \quad k_i \in \{1, \dots, 4\}. \quad (2)$$

- Why the lagged weights? Next two slides.

The Instrument Is a Shift-Share Design

Two maturity buckets whose par shares sum to one, so the frozen aggregation decomposes exactly:

$$\widetilde{MIV}_{it}^{firm} = \underbrace{\log \frac{A_{2t}}{N_{2t}}}_{c_t} + \underbrace{\frac{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k} \mathbb{1}\{TTM_{jt} < 10\}}{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k}}}_{\tilde{s}_{it}} \times \underbrace{\left(\log \frac{A_{1t}}{N_{1t}} - \log \frac{A_{2t}}{N_{2t}} \right)}_{g_t}$$

- **Share** $\tilde{s}_{it}$: firm i 's par share of bonds with time to maturity (TTM) below 10 years, on the portfolio frozen at $t - k$: *exposure* to the mandate boundary
- **Shift** g_t : log ratio of mutual-fund assets per bond, short vs. long bucket: common to all firms
- c_t : long-bucket level, absorbed by **time fixed effects (FE)**; permanent part of $\tilde{s}_i$ by **firm FE**

Two buckets, shares sum to one $\Rightarrow$ **two-industry case** of Goldsmith-Pinkham et al (2020): the instrument *is* the share, rescaled by the shift: **the share carries the identification**.

Identifying assumption: within-firm moves in $\tilde{s}_{it}$ exogenous to outcome *innovations*, given controls.

Making the Share Predetermined: The Frozen Portfolio

Within-firm movement in a firm's short-maturity share has two sources:

- **Mechanical aging:** bonds cross the 10-year line on indenture-fixed dates; predetermined
- **Active maturity management:** issuance, buybacks, exchanges; endogenous

Today's portfolio carries both; the frozen portfolio carries only the scheduled crossings:

$$s_{it} = \frac{\sum_{j \in \mathcal{B}_{i,t}} \text{AmtOut}_{j,t} \mathbb{1}\{TTM_{jt} < 10\}}{\sum_{j \in \mathcal{B}_{i,t}} \text{AmtOut}_{j,t}} \quad \text{vs.} \quad \tilde{s}_{it} = \frac{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k} \mathbb{1}\{TTM_{jt} < 10\}}{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k}}$$

Same formula; the subscript change $t \rightarrow t - k$ removes the firm's discretion since $t - k$.

Timing: portfolio frozen at $t - k \rightarrow$ aged deterministically to $t \rightarrow$ ownership at $t \rightarrow$ outcomes at $t + 1$. $k_i \in \{1, \dots, 4\}$ set by data coverage, constant within firm (level absorbed by firm FE).

Checks: first-stage $F = 30\text{--}36$; insurer placebo; GPSS diagnostics.

freeze-horizon details

Mutual Fund Ownership Amplifies the Liquidity Effect

$$\Delta \text{Employment}_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Regression Fund Category	OLS	OLS Mutual	IV Mutual
Liquidity Premium	-0.5894*** (-2.82)	-0.4554** (-2.21)	-0.3440* (-1.76)
Bond Ownership		0.0061 (0.41)	0.1530 (1.23)
Liquidity Premium $\times$ Bond Ownership		-1.1284** (-2.02)	-1.8580** (-2.37)
No. of Observations	5,924	5,924	5,924
First-stage F-stat			34.7
Firm Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Clustering	Firm	Firm	Firm

A 1-SD rise in the liquidity premium lowers employment growth by 9.3% of a SD. Moving from the 25th to the 75th percentile of mutual fund ownership amplifies the response by a further 4.7% of a SD — **about half of the baseline effect on top of it.**

Placebo: Insurance Fund Ownership

Placebo tests using insurance fund bond ownership:

- Literature shows that insurance funds are more stable, longer-horizon bond holders (Li, 2025; Zhou, 2025).
- Insurance fund bond ownership does not exhibit discontinuity regarding maturity.
- We use insurance fund bond ownership in the place of mutual fund bond ownership, and conduct the placebo tests.

Placebo Results: No Effect for Insurance Ownership

$$\Delta \text{Employment}_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} \\ + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Regression Fund Category	OLS	OLS Mutual	IV Mutual	OLS Insurance	IV Insurance
Liquidity Premium	-0.5894*** (-2.82)	-0.4554** (-2.21)	-0.3440* (-1.76)	-0.5843*** (-3.06)	-1.8441 (-0.79)
Bond Ownership		0.0061 (0.41)	0.1530 (1.23)	0.0032 (0.31)	-0.2162 (-0.92)
Liquidity Premium $\times$ Bond Ownership		-1.1284** (-2.02)	-1.8580** (-2.37)	0.0981 (0.22)	9.1983 (0.56)
No. of Observations	5,924	5,924	5,924	5,414	5,414
First-stage F-stat			34.7		13.2
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm

Takeaway: the insurance interaction is never significant (and its instrument is much weaker) — the amplification is specific to **run-prone** holders, not to bond ownership concentration per se.

Same Pattern for Investment and Debt Issuance

$$Y_{i,t} = \beta_1 \text{Liquidity Premium}_{i,t-1} \times \text{Bond Ownership}_{i,t-1} \\ + \beta_2 \text{Liquidity Premium}_{i,t-1} + \beta_3 \text{Bond Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Dependent Variable	Investment			Debt Issuance		
Fund Category Regression	Mutual OLS	Mutual IV	Insur. OLS	Mutual OLS	Mutual IV	Insur. OLS
Liquidity Premium	-0.0241 (-0.51)	-0.0688 (-1.16)	-0.1484* (-1.69)	-0.0280 (-0.10)	0.2157 (0.66)	-0.2926 (-0.86)
Bond Ownership	-0.0066 (-0.76)	-0.1453** (-1.98)	-0.0011 (-0.13)	-0.0107 (-0.80)	-0.0361 (-0.44)	0.0240** (1.98)
Liquidity Premium × Bond Ownership	-0.7286*** (-2.61)	-0.9394** (-2.03)	0.3168 (0.96)	-1.9663*** (-3.75)	-3.9332*** (-3.24)	0.0666 (0.13)
First-stage F-stat		31.6			36.0	
No. of Observations	5,957	5,957	5,443	5,968	5,968	5,452
Firm & Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Takeaway: the pattern replicates for investment and, most strongly, for debt issuance (p75 vs. p25 ownership: a further 7.1%-SD lower issuance per 1-SD of liquidity risk); insurance interactions never negative and significant.

The Liquidity Interaction Is Not Proxying for Other Channels

$$\Delta \text{Emp}_{i,t} = \beta_1 \text{Liq. Premium}_{i,t-1} \times \text{Ownership}_{i,t-1} + \beta_2 \text{Channel}_{i,t-1} \times \text{Ownership}_{i,t-1} \\ + \beta_3 \text{Liq. Premium}_{i,t-1} + \beta_4 \text{Ownership}_{i,t-1} + \mathbf{X}'_{i,t-1} \boldsymbol{\beta} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

Dependent Variable	Δ Employment		
Competing channel	Fund Flow Shock	Rollover Risk	Cash Saving
Liquidity Premium	-0.3751* (-1.83)	-0.4506* (-1.77)	-0.4262* (-1.71)
Liquidity Premium × Bond Ownership	-1.7310*** (-2.86)	-1.4320* (-1.81)	-1.3570* (-1.78)
Channel × Bond Ownership	0.0047* (1.92)	-0.4210* (-1.81)	0.0774 (0.20)
First-stage F-stat	30.40	13.27	29.32
No. of Observations	5,816	5,922	5,922
Firm & Time FE	Yes	Yes	Yes

Takeaway: β_1 remains negative and significant when each competing interaction is added. Similar for investment and debt issuance.

All outcomes

Roadmap

- ① Data and Empirical Results
- ② Model: Firms and an OTC Bond Market
- ③ Conclusion

Model Overview

Goal: quantify how bond market liquidity and investor composition shape employment.

Environment: discrete time, monthly; heterogeneous firms + bond investors

- Firms can issue **long-duration defaultable bonds** and choose investment, hiring, new bond issuance; costly equity issuance; endogenous default.
- Bonds are held by investors who trade in an **OTC secondary market** with search frictions and face **distress shocks** (Duffie–Gârleanu–Pedersen 2005; He–Milbradt 2014)
- Aggregate **liquidity state** l_t moves distress risk, dealer matching, holding costs, recovery.
- Liquidity reaches the firm through the **issue price** of new bonds: issuance proceeds move one-for-one with bond-market liquidity.

Today: one investor class; compare two economies differing in investor **risk capacity** κ (mutual-fund-like vs. insurer-like demand).

Full model (in progress): heterogeneous investors trade with each other.

Firms: Technology and Long-Duration Debt

Technology (as before): $Y_{it} = Z_{it}X_t(K_{it}^\alpha N_{it}^{1-\alpha})^\theta$, convex adjustment costs on investment and hiring, predetermined labor, rigid wage $W_t = \tau_1 e^{\tau_2 X_t}$.

Debt: exponentially maturing perpetuity (Leland 1998):

$$B_{i,t+1} = (1 - m)B_{it} + b_{it}, \quad b_{it} \geq 0,$$

each unit matures w.p. $m = 1/84$ per month, pays coupon c ; no buybacks $\Rightarrow$ deleveraging only through maturities.

Default: limited liability, taste-shock smoothing

taste shocks

$$\mathcal{D}_{it} = \frac{1}{1 + e^{V^r(S_{it})/\xi_d}}, \quad S_{it} = (K, N, B, x, l, z)_{it} \text{ — six states}$$

Recovery: $\varphi(K, B) = \min\{\varphi_k(1 - \delta)K/B, \varphi_b\}$

The Firm's Dynamic Problem

Bellman (conditional on continuing):

$$V^r(S_{it}) = \max_{K', N', b \geq 0} \left\{ D_{it} + \mathbb{E}_t [M_{t,t+1} V(S_{i,t+1})] \right\},$$

$$E_{it} = (1 - \tau)(Y_{it} - W_t N_{it} - f) + \tau \delta K_{it} - I_{it} - \Phi_{it} - (m + (1 - \tau)c)B_{it} + p_{it}^P b_{it}$$

Liquidity price enters the budget: the **issue price** p^P , which comes from the OTC investor block

SDF (Berk et al. 1999; Zhang 2005), loading on productivity *and* liquidity risk:

$$M_{t,t+1} = \beta \frac{e^{\gamma_x(x_t - x_{t+1}) + \gamma_l(l_t - l_{t+1})}}{\mathbb{E}_t [e^{\gamma_x(x_t - x_{t+1}) + \gamma_l(l_t - l_{t+1})}]}$$

The OTC Secondary Market

Investors hold 0 or 1 unit; each period a holder is **non-distressed** (H) or **distressed** (L):

- Distress shock w.p. $\eta(l_t)$: a sudden reason to sell.
- A distressed holder must **find a dealer**: succeeds w.p. $\lambda(l_t)$; Nash bargaining with seller weight β_N :

$$p_{it}^A = V_{it}^H, \quad p_{it}^B = \beta_N V_{it}^H + (1 - \beta_N) V_{it}^L, \quad \text{bid-ask} = (1 - \beta_N) \Delta V_{it}$$

- Failure to sell: holding cost $h(l_t)$, and carrying the recovery disadvantage $\Delta\varphi_{it}$.
- **Closed form** for the liquidity wedge (fresh-draw distress):

$$\Delta V_{it} = \frac{(1 - \lambda(l_t))(h_{it} + \Delta\varphi_{it})}{1 - \lambda(l_t)(1 - \beta_N)} \quad \text{— distress is most costly exactly when it is most likely.}$$

Investor Values Are Endogenous: the Holder's Bellman Equation

A non-distressed holder's value per unit of face, at the firm's post-decision state $y = (K', N', B')$:

$$V_{it}^H = \mathbb{E}_t \left[M_{t,t+1} \left\{ \underbrace{(c + m + (1 - m)[(1 - \eta')V_{i,t+1}^H + \eta'V_{i,t+1}^L])}_{\text{coupon + principal + continuation, if firm survives}} (1 - \mathcal{D}') + \underbrace{\varphi^H(K', B', I') \mathcal{D}'}_{\text{recovery if default}} \right\} \right]$$

A distressed holder sells at the bid w.p. λ , else pays h and carries the recovery disadvantage:

$$V_{it}^L = \lambda p_{it}^B + (1 - \lambda)(V_{it}^H - h_{it} - \Delta\varphi_{it}) \implies \Delta V_{it} = \frac{(1 - \lambda)(h_{it} + \Delta\varphi_{it})}{1 - \lambda(1 - \beta_N)}$$

(substituting the Nash bid and solving the within-period fixed point).

- **Equilibrium** = the joint fixed point of $(V^r, \text{policies}, V^H)$: V^H depends on the firm's **default risk** $\mathcal{D}'$ and **future policies** (dilution — the hard part of long-duration debt), and the firm's problem depends on V^H through the issue price.
- Everything liquidity-related that follows (p^P , bid-ask, the liquidity premium) is built from (V^H, V^L) .

The Primary Market: Where Liquidity Meets the Firm

New bonds are sold *before* the distress draw; a buyer knows it faces this period's draw:

$$p_{it}^P = V_{it}^H - \eta(l_t) \Delta V_{it} - \kappa b_{it}$$

- $V^H - \eta \Delta V$: the holder's **endogenous Bellman value** (previous slide) at the firm's chosen (K', N', B') , net of expected distress cost — the schedule internalizes dilution and default risk.
- κ : **investor risk capacity** — demand slope from a quadratic cost of concentrating exposure in one issuer (VaR limits, mandates, balance-sheet space): **low** κ = deep, insurer-like demand; **high** κ = shallow, mutual-fund-like demand.

Transmission chain — two legs, one liquidity state:

$$l_t \downarrow \Rightarrow \eta \uparrow, \lambda \downarrow, h \uparrow \Rightarrow p^P \downarrow \Rightarrow \text{issuance proceeds} \downarrow \Rightarrow \text{internal funds} \downarrow \Rightarrow \text{hiring, investment} \downarrow$$

Calibration

Param.	Interpretation	Value	Param.	Interpretation	Value
β	Discount factor	0.9959	m	Maturity hazard ($1/m = 7y$)	1/84
τ	Corporate tax	0.30	c	Coupon ($= r_f$, par)	0.0042
θ	Returns to scale	0.82	$\eta(l)$	Distress hazard (mean / bad l)	0.10 / 0.29
α	Capital share	0.28	$\lambda(l)$	Dealer matching (mean / bad l)	0.35 / 0.13
δ, δ_n	Depreciation, separations	0.01, 0.021	β_N	Seller bargaining weight	0.50
c_k, c_n	Adjustment costs	2.0, 2.0	h_0	Holding cost (level, l -elastic)	0.012
f	Fixed operating cost	0.10	φ_k, φ_b	Recovery, cap	0.5, 0.5
ψ_1	Equity issuance cost	0.05	$c_{\varphi,1}$	Distressed-holder recovery	0.45
ξ_d	Taste-shock scale	0.10	κ	Risk capacity: insurer / MF	0.02 / 0.1
ρ_x, σ_x	Aggr. productivity	0.95, 0.10	ρ_l, σ_l	Liquidity state	0.95, 0.02
$\rho_z, \sigma_z, \bar{z}$	Idio. productivity	0.97, 0.10, -2.6	γ_x, γ_l	Prices of risk	0.75, 0.75

Monthly. Firm block: Belo–Favilukis–Lin 2025, Hennessy–Whited 2007, live-default benchmark of the one-period model.

Solution Method: Two Approaches

Six-dimensional state, kinked nonconcave objective, bond price depends on value and policies.

- ❶ **Grid-based solver** (JAX, GPU): exact fixed point of the discretized problem
 - global candidate search ranked by the Bellman objective (no FOCs);
 - joint value–price sweeps.
- ❷ **NN solver** (actor–critic): firm value + investor-value critics, firm policy actor; prices computed in closed form from the critic.

NN details

Deep Learning for the OTC Model

Six-dim. state, kinked nonconcave objective, price recursive in future policies. Three networks:

$$V^r(S) \approx \hat{V}^r(S; \Theta_V), \quad V^H(y; S) \approx \hat{V}^H(y; \Theta_H), \quad (K', N', b) \approx \hat{\pi}(S; \Theta_\pi)$$

- **Critics:** firm value $\hat{V}^r$; investor value $\hat{V}^H$ on post-decision points, sigmoid head scaled by the par bound $(c + m)/(r_f + m)$.
- **Actor:** multiplicative heads for K', N' ; issuance head a scaled softplus $\Rightarrow b \geq 0$ holds by construction, $B' = (1 - m)B + b$.
- **Close-form whenever possible:** $p^P, \Delta V$, bid/ask follow in closed form from the target critic $\hat{V}^H$ — gradients flow through the price schedule.

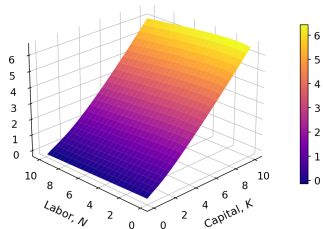
At each step: sample states (oversampling the default and low- l corners), form

$$L = \underbrace{(\hat{V}^r - D - \mathbb{E}[M\hat{V}'])^2}_{\text{Bellman residual}} + \underbrace{(\hat{V}^H - \text{RHS of the } V^H \text{ recursion})^2}_{\text{investor residual}} - \underbrace{(D + \mathbb{E}[M\hat{V}'])}_{\text{actor objective (ascent)}},$$

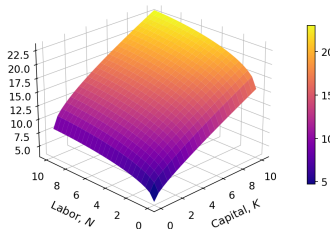
Optimization with SGD variant; no first-order conditions are used (kinks), FOCs serve as diagnostics only.

Results (I): Firm Value and Default

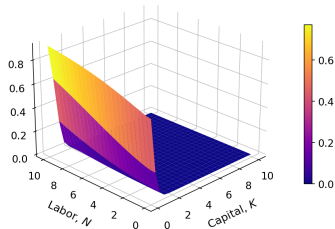
$V^r, z = -3.61$ (low)



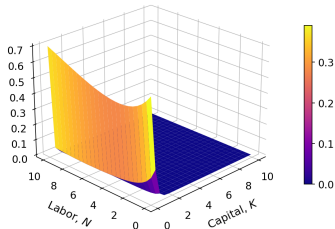
$V^r, z = -1.59$ (high)



$D, z = -3.61$ (low)



$D, z = -2.60$ (mid)



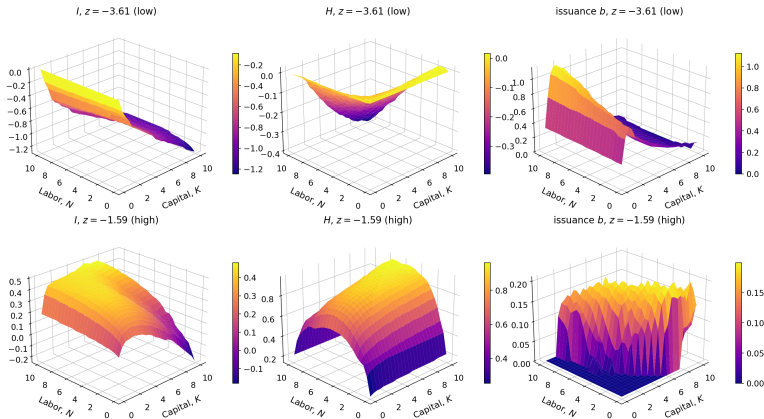
V^r and default probability D over (K, N) at middle B , mean I .

V^r rises in K , is concave in N (rigid wage + adjustment costs).

Default survives only where capital is small relative to the wage bill and debt: with 7-year maturity, flow service per unit of face is ~ 0.015 , not ~ 1 — **long maturity defuses rollover**.

At low z the default region is wide; at mid z it retreats to the smallest firms.

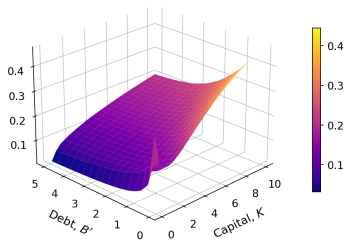
Results (II): Firm Policies



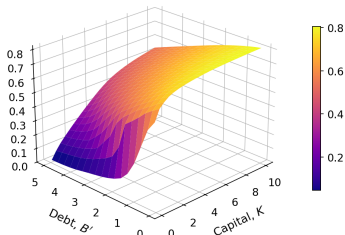
- High z (bottom): investment and hiring hump-shaped (convex adjustment costs); issuance small and positive — on the order of the **maturing flow** ($B' \approx B$: firms roll over).
- Low z (top): survivors **disinvest and shed labor**; issuance concentrates in a band near the default boundary (funding the wage bill through distress).

Results (III): Investor Valuations and the Issue Price

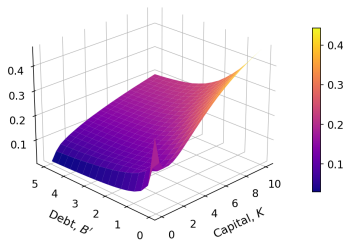
$V^H, z = -3.61$ (low)



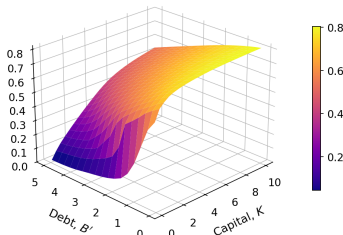
$V^H, z = -2.60$ (mid)



$p^P, z = -3.61$ (low)



$p^P, z = -2.60$ (mid)

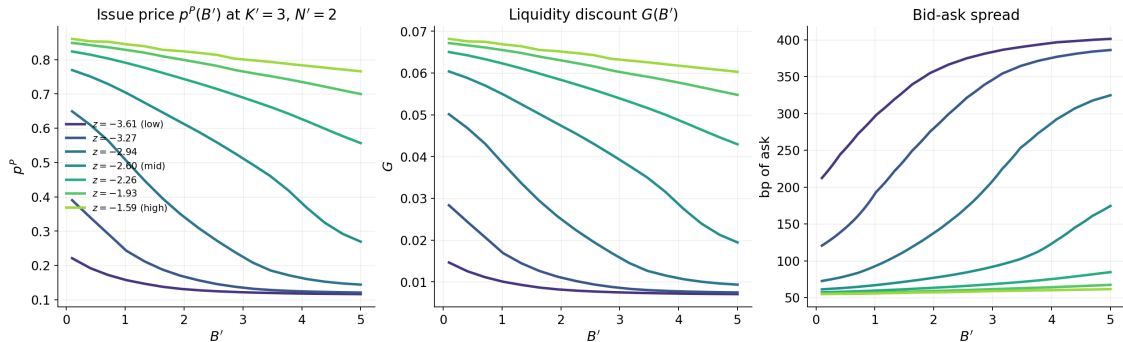


Holder value V^H and issue price p^P per unit of face, over (K', B') at mean l .

Prices rise with collateral K' and **fall in the face value that must share it** — the price schedule internalizes dilution (leverage ratchet).

Surfaces steepen exactly where default risk concentrates: the credit–liquidity interaction the empirics exploit.

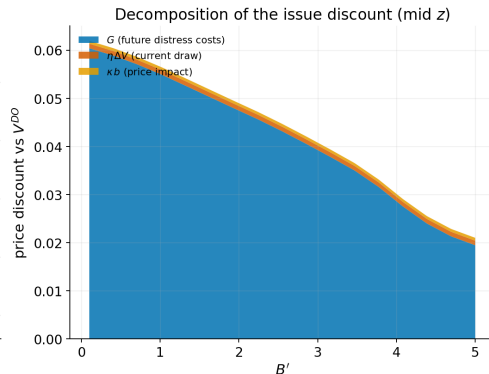
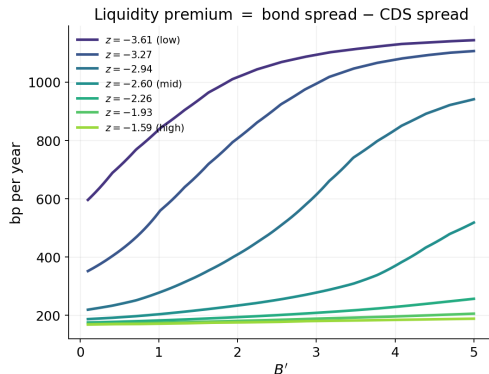
Results (IV): Bond Prices, Liquidity Discount, Bid-Ask



- Price schedules slope down in issuance (dilution) and collapse for risky firms.
- **Bid-ask: $\sim 55\text{bp}$ (safe) to $\sim 400\text{bp}$ (near default).**
- Liquidity discount G is larger for *safer* bonds: they survive longer and accrue distress costs over the full 7-year life (duration effect).

Solved one-class model, main calibration, mean liquidity state.

Results (V): The Liquidity Premium Across Firms



- LP = bond spread – CDS spread: **positive everywhere, firm-specific, steeply increasing in default risk** — the cross-sectional regressor of the empirical part, as an equilibrium object.
- Decomposition: capitalized *future* distress costs G dominate; price impact κb and the current draw add on top.

LP measurement: model = data

Solved one-class model, main calibration, mean liquidity state.

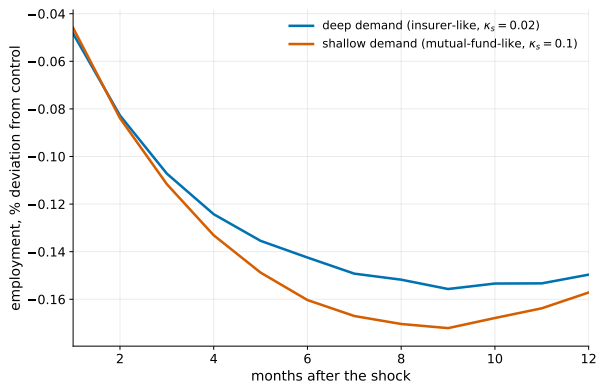
Results (VI): Investor Risk Capacity κ — Deep vs. Shallow Demand

Two economies identical except the demand slope κ :

Ergodic moment	Deep demand ($\kappa = 0.02$) <i>insurer-like</i>	Shallow demand ($\kappa = 0.1$) <i>mutual-fund-like</i>
Book leverage B' / K'	0.18	0.15
Default rate (mature, ann.)	4.2%	4.2%
Credit spread (ann., mid)	11.7%	9.7%
Liquidity premium (ann.)	242 bp	216 bp
Bid-ask spread	87 bp	83 bp
Issue concession	46 bp	66 bp
Issuance frequency (monthly)	0.94	0.95
Mean employment	7.86	7.84

- Deep demand supports **more leverage**; the extra debt endogenously *raises* spreads and the liquidity premium.
- Shallow demand pays a **larger concession per issue** (66 vs. 46 bp): the price impact investors charge.
- Steady-state employment nearly **identical**: capital structure absorbs the difference — the investor base matters for the *response to shocks* $\Rightarrow$ next slide.

Results (VII): Employment Response to a Liquidity Shock, by Investor Base



- **A real recession:** employment falls for 3 quarters after an adverse 2-sd liquidity shock (η : 10% $\rightarrow$ 29%; working-capital spread 52bp $\rightarrow$ 1.5%/mo), trough at month 9, recovery afterwards.
- **The investor base matters:** same shock, $\sim 10\%$ more job losses for mutual fund held firms

Roadmap

- ① Data and Empirical Results
- ② Model: Firms and an OTC Bond Market
- ③ Conclusion

Conclusion

- Bond market liquidity risk predicts lower firm employment growth, investment, and debt issuance.
- **Mutual fund ownership amplifies** the employment effect by about **half** of the baseline (p75 vs. p25 ownership), causally identified with a frozen-portfolio mandate IV.
- An OTC model of the bond market delivers the mechanism: distress shocks and search frictions create firm-specific liquidity premia that pass through the **issue price** into internal funds and hiring; a liquidity shock causes a three-quarter employment recession whose **depth is set by investor risk capacity** κ ($\sim 10\%$ deeper under mutual-fund-like demand).

Next steps:

- Calibrate the liquidity block formally to bid-ask, turnover, and liquidity-premium moments; NN solver at the main calibration.
- **Two investor classes** (mutual funds and insurers) trading with each other: investor composition as a firm-level state, endogenous composition dynamics.

Literature

- Interaction of corporate bonds liquidity and default (He and Milbradt (2014), Chen et al. (2018), Chang (2023))
- Impact of financial frictions (excess bond premium) on real activities (Gilchrist and Zakrajšek (2012))
- Impact of labor market frictions on credit markets (Favilukis, Lin, and Zhao (2020))
- Role of institutional investor in shock transmission: Li and Yu (2021), Coppola (2025 RFS), Fang (2023), Zhou (2024), Li (2024)

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Appendix: Shift-Share Design to Refine Firm-level MIV

Identification rests on within-firm movement in the frozen short-maturity share $\tilde{s}_{it}$ of

$\widetilde{MIV}_{it}^{firm} = c_t + \tilde{s}_{it} g_t$, which has two sources:

- **mechanical aging** of outstanding bonds across the 10-year line: predetermined;
- **active maturity management**: endogenous, stressed firms shorten maturity.

Fix: freeze the bond portfolio. Take each firm's bonds as of year $t - k$ (par amounts, contractual maturities), age them *deterministically* to t , and rebuild the share from the frozen portfolio:

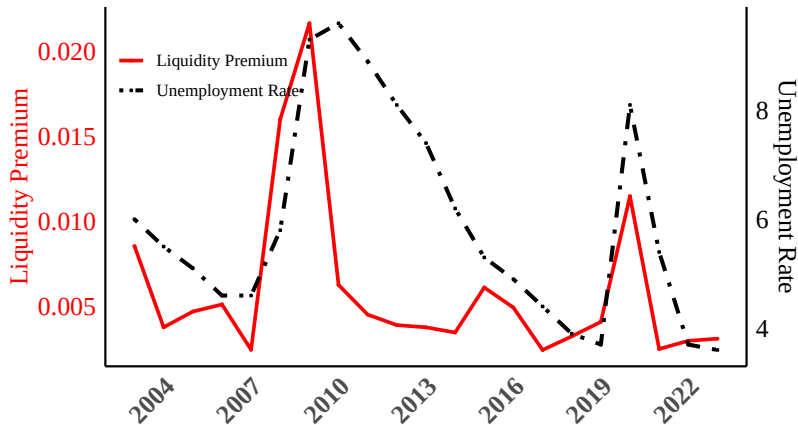
$$\tilde{s}_{it} = \frac{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k} \mathbb{1}\{TTM_{jt} < 10\}}{\sum_{j \in \mathcal{B}_{i,t-k}} \text{AmtOut}_{j,t-k}}, \quad \widetilde{MIV}_{it}^{firm} = c_t + \tilde{s}_{it} g_t$$

Freeze horizon: $k_i = \max\{1, \min\{4, 2002 - \tau_i\}\}$, τ_i the first year of firm i 's eMAXX bond coverage (data from 1998, sample from 2002); constant within firm, so its level is absorbed by firm fixed effects.

Timing: frozen at $t - k \rightarrow$ deterministic aging $\rightarrow$ ownership at $t \rightarrow$ outcomes at $t + 1$: only pre-set maturity structure moves the instrument.

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Aggregate Results: Liquidity Risk vs Unemployment Rate



Corporate bond liquidity risk vs unemployment rate: correlation = 0.49. [back](#)

Aggregate-Level Liquidity Risk vs Labor Market Dynamics

$$Y_t = \beta_1 \text{Liquidity Premium}_t + \mathbf{X}_t' \boldsymbol{\beta} + \varepsilon_t$$

Dependent Variable	Labor Market Tightness	Unemployed Persons Per Job Opening Ratio	Unemployment Rate	Employment Growth
Panel A				
Bid-Ask Spread	-0.8935* (-1.83)	2.9728*** (2.91)	2.9661** (2.26)	-0.0037* (-1.68)
Panel B				
Aggregate Liquidity Premium	-38.2147** (-2.04)	102.1171** (2.39)	95.7323* (1.73)	-0.2819*** (-2.73)
HAC Robust SE	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes

$\mathbf{X}_t'$: industry production growth, ΔVIX .

Liquidity Risk, Financial Constraints, and Economic Condition

Dependent Variable	Δ Employment		
	(1)	(2)	(3)
Liquidity Premium	-0.5973** (-2.47)	-2.1384** (-2.38)	-1.7601* (-1.93)
Financial Constraint		-0.0109 (-0.45)	-0.0083 (-0.34)
Liquidity Premium $\times$ Financial Constraints		-0.3674* (-1.84)	-0.3669* (-1.80)
Liquidity Premium $\times$ $\mathbb{1}(\text{Recession})$			-3.1134* (-1.71)
Liquidity Premium $\times$ Financial Constraints $\times$ $\mathbb{1}(\text{Recession})$			-0.7048* (-1.74)
No. of Observations	5,964	5,964	5,964
Firm Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Clustering	Firm	Firm	Firm

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Robustness: 10-Year Tenor CDS Contracts

Dependent Variable	Δ Employment					Debt Issuance					Investment				
	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV
Liquidity Premium	-0.4583*	-0.3038	-0.3149	-0.4601*	-0.9364	-0.3401***	-0.2455**	-0.2485**	-0.2255**	-0.8296*	-0.1802**	-0.0411	0.0537	-0.3214***	-0.5096
	(-1.89)	(-1.21)	(-1.24)	(-1.86)	(-1.32)	(-4.09)	(-2.34)	(-2.08)	(-2.24)	(-1.72)	(-2.07)	(-0.47)	(0.42)	(-2.84)	(-1.15)
Bond Ownership		0.0070	-0.0621	0.0026	-0.1815		-0.0346**	-0.0772*	0.0254**	-0.2100		-0.0079	-0.0231	-0.0016	-0.1401
		(0.56)	(-1.01)	(0.32)	(-1.13)		(-2.33)	(-1.86)	(2.07)	(-0.87)		(-0.90)	(-1.11)	(-0.24)	(-1.23)
Liquidity Premium × Bond Ownership		-1.256**	-1.371*	0.2497	3.708		-0.6238*	-0.6911*	0.4663	5.693		-0.9762**	-1.685**	0.6730**	1.588
		(-2.30)	(-1.85)	(0.61)	(0.77)		(-1.85)	(-1.67)	(0.93)	(1.37)		(-2.19)	(-2.15)	(2.06)	(0.65)
Channel		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance
No. of Observations	5,796	5,796	5,796	5,317	5,317	5,941	5,941	5,941	5,434	5,434	5,827	5,827	5,827	5,344	5,344
First-stage F -stat			39.70		0.52			172.31		1.62			82.83		1.31
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Robustness: Excluding Crisis Periods

Dependent Variable	Δ Employment					Debt Issuance					Investment				
	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV	OLS	OLS	IV	OLS	IV
Liquidity Premium (LP)	-0.3949 (-1.42)	-0.2375 (-0.89)	-0.2273 (-0.86)	-0.4403 (-1.62)	-1.469 (-0.76)	-0.5280*** (-3.21)	-0.3076** (-2.10)	-0.2201 (-0.95)	-0.5704* (-1.96)	-2.806 (-0.65)	-0.1518* (-1.74)	-0.0895 (-0.87)	0.1315 (0.87)	-0.2346** (-2.08)	-0.6966 (-1.43)
Bond Ownership		0.0094 (0.72)	-0.0149 (-0.27)	0.0024 (0.27)	-0.0765 (-0.33)		-0.0287** (-2.01)	-0.0754* (-1.79)	0.0234* (1.65)	0.4376 (0.32)		-0.0053 (-0.58)	-0.0082 (-0.40)	-0.0001 (-0.07)	-0.1174 (-1.09)
Liquidity Premium × Bond Ownership		-1.4940** (-2.36)	-1.6480* (-1.87)	0.2756 (0.54)	8.8430 (0.64)		-1.1880** (-2.36)	-1.6900** (-2.00)	1.5660** (1.99)	25.4850 (0.55)		-0.4505 (-0.92)	-1.9750** (-2.23)	0.2142 (0.66)	2.5850 (0.99)
Channel		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance		Mutual	Mutual	Insurance	Insurance
No. of Observations	5,064	5,064	5,064	4,574	4,574	5,184	5,184	5,184	4,663	4,663	5,092	5,092	5,092	4,598	4,598
First-stage F -stat			34.73		0.22			131.93		0.14			71.23		1.28
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Appendix: The Model Prices the Empirical Liquidity Premium

Default-only value V^{DO} : the same bond to a holder who never gets distressed
 $\Rightarrow$ the model's **CDS-implied value** (default risk only).

Spreads, exactly as measured in the data (bond–CDS basis, Longstaff–Mithal–Neis 2005):

$$\text{LP}_{it} = \underbrace{\frac{c+m}{p_{it}} - m - r_f}_{\text{bond spread}} - \underbrace{\left(\frac{c+m}{V_{it}^{\text{DO}}} - m - r_f \right)}_{\text{CDS spread}} \propto \underbrace{G_{it}}_{\text{liquidity discount}} + \underbrace{\eta \Delta V_{it}}_{\text{distress exposure}} + \underbrace{\kappa b_{it}}_{\text{price impact}}$$

$G = V^{\text{DO}} - V^H$ = expected PV of **future distress costs** over the bond's remaining life:

$$G_{it} = \mathbb{E}_t \left[M_{t,t+1} (1-m)(1-\mathcal{D}') (G_{i,t+1} + \eta' \Delta V_{i,t+1}) \right]$$

$\Rightarrow$ the regressor of the empirical part is an equilibrium object of the model:
liquidity premia are **firm-specific**, countercyclical, and rise with default risk.

Appendix: Model vs. Data Moments

Moment	Data	Model
<i>Panel A: Firm-level moments</i>		
ΔDebt	0.038	0.004
$\text{SD}(\Delta\text{Debt})$	0.213	0.120
$\text{AR1}(\Delta\text{Debt})$	-0.126	0.867
$\Delta\text{Employment}$	0.010	0.005
$\text{SD}(\Delta\text{Employment})$	0.134	0.063
$\text{AR1}(\Delta\text{Employment})$	-0.011	0.974
Investment Rate	0.194	0.170
$\text{SD}(\text{Investment Rate})$	0.115	0.396
$\text{AR1}(\text{Investment Rate})$	0.307	0.982
<i>Panel B: Aggregate moments</i>		
Leverage Ratio	0.328	0.485
Default Probability	0.017	0.013
Credit Spread	0.011	0.017
Stock Return	0.089	0.120
Stock Return Volatility	0.156	0.142
Sharpe Ratio	0.581	0.483

Takeaway: default rate and leverage close; credit spread realistic; firm-level dynamics same order of magnitude. Persistence too high; investment-rate vol too high.

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Approximate V^r and Policies by Neural Network (Feed-Forward, Fully Connected)

- Let $\omega = (K, N, B, X, Z)$. We approximate each of $\widehat{V}^r, \widehat{K}, \widehat{N}, \widehat{B}$ by an NN of the form

$$\mathbf{h}^{(1)} = \phi^{(1)}(\mathbf{W}^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(\mathbf{W}^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer } H$$

$$\text{output} = \sigma(\mathbf{h}^{(H)}) \quad \dots \widehat{V}^r, \widehat{K}, \widehat{N}, \text{ or } \widehat{B}$$

- Terminology:
 - H : number of *hidden layers*
 - Length of $\mathbf{h}^{(i)}$: number of *neurons* per hidden layer
 - $\phi^{(i)}$: *activation* for hidden layer i
 - σ : *activation* for the output layer
 - $\Theta = (\mathbf{W}^{(1)}, \dots, \mathbf{W}^{(H)}, \mathbf{b}^{(1)}, \dots, \mathbf{b}^{(H)})$: *parameters*, trained by SGD.

Appendix: Taste-Shock Smoothing of the Default Decision

Why: discrete default $\Rightarrow$ non-differentiable value $\Rightarrow$ breaks gradient-based deep-learning solution. Standard fix from sovereign-default literature (Dvorkin–Saprizza–Sosa-Padilla 2021; Chatterjee et al. 2023).

Setup. Add i.i.d. type-I EV shocks ϵ^r, ϵ^d with scale $\xi_d > 0$ to continuation value V^r and default value V^d . The ex-ante firm value and default probability are:

$$V = \mathbb{E}_\epsilon \max\{V^r + \epsilon^r, V^d + \epsilon^d\} = \xi_d \log \left[\exp \frac{V^r}{\xi_d} + \exp \frac{V^d}{\xi_d} \right], \quad \mathcal{D} = \frac{\exp(V^d/\xi_d)}{\exp(V^r/\xi_d) + \exp(V^d/\xi_d)}$$

Normalize $V^d = 0$:

$$V = \xi_d \log \left[1 + \exp \frac{V^r}{\xi_d} \right], \quad \mathcal{D} = \frac{1}{1 + \exp(V^r/\xi_d)}$$

Deterministic limit. As $\xi_d \rightarrow 0$: $\mathcal{D} \rightarrow \mathbf{1}_{\{V^r < 0\}}$, recovering the standard limited-liability default rule.

Calibration: $\xi_d = 0.10$.

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